

MEMORANDUM

TO: Chief Underwriting Officer
FROM: Hari Acharya, Analytics
DATE: May 20, 2026
RE: Auto Book Loss Performance — Findings and Recommended Actions

EXECUTIVE SUMMARY

Our auto book is currently running at a **57.8% loss ratio**, technically profitable and below the 65% industry benchmark. However, this aggregate number masks dangerous concentration: just **0.04% of policies (249 of 678,013) generate 38% of all claim losses**, and a single demographic segment — drivers aged 18-25 — runs a **173.8% loss ratio**, more than three times the book average. Without targeted intervention, these concentrations create asymmetric downside: a small shock in any of these segments would erode the margin rapidly. I recommend three discrete actions projected to improve underwriting profit by **€21.4M over twelve months** while affecting under 5% of book volume.

KEY FINDINGS

Finding 1: A single age segment drives disproportionate losses.

Drivers aged 18-25 represent 4.5% of book exposure but generate a **173.8% loss ratio**. The decomposition is unusual: this segment is broken on both axes — frequency is 74% above the book average (**0.175 vs 0.101** claims per policy-year) AND severity is 161% above (**€4,332 vs €1,659** per claim). Every other driver age bucket sits between 46% and 55% loss ratio. This is not a market-wide effect; it is a segment-specific underwriting problem.

Finding 2: Two segments are severity-driven, not frequency-driven.

Area B (73.4% loss ratio) and vehicle ages 11+ (67.0% loss ratio) both run claim frequencies *below* the book average, but their average claim costs are 40-50% above book. This distinction matters operationally: re-pricing these segments will not solve the problem — when claims are rare but expensive, the cost is in the type of risk written, not the volume. Underwriting tightening is the appropriate response.

Finding 3: Loss concentration is extreme — 0.04% of policies = 38% of losses.

Sorting policies by total claim amount, the worst **249 policies (0.04% of the book) account for 37.9% of all claim dollars**. Their combined premium contribution was €42,000 — meaning these policies ran a 42,000%+ loss ratio. This level of concentration suggests identifiable individual risk, not random variation, and is the strongest argument for selective non-renewal at the next renewal cycle.

RECOMMENDATIONS

Action 1: Re-price drivers aged 18-25 by 40%.

Projected to reduce the segment's effective book share through natural attrition (assume 60% retention) while increasing premium per retained policy. Combined effect: book-level loss ratio improvement of **4.2 points**. Net premium impact is slightly negative (–€1.1M) but losses fall by €4.9M.

Action 2: Tighten underwriting in Area B.

Selectively non-write the highest-risk 25% of new business in this geography. Because losses are concentrated in the worst risks, this should remove approximately 40% of segment losses while reducing premium by only 25%. Incremental loss ratio improvement: **1.2 points** beyond Action 1.

Action 3: Non-renew the top 249 policies by loss history.

These policies generate €17.9M in losses against €42,000 in premium — a structural mispricing rather than a statistical fluke. Non-renewal at the next cycle removes the losses entirely with negligible premium impact. Incremental loss ratio improvement: **17.9 points** beyond Actions 1 and 2.

Combined projected impact:

Loss ratio improves from 57.8% to **34.5%** (a **23.3-point** improvement). Underwriting profit grows from €43.7M to €65.1M — a **€21.4M** twelve-month gain. Book volume reduction is less than 5%.

ASSUMPTIONS AND LIMITATIONS

Premium was not provided in the source data and was approximated using the BonusMalus column (industry-standard French market practice) on a €500 base rate. Different base rate assumptions would shift the absolute loss ratio but not the relative findings between segments. Retention elasticity is assumed at 60% for 18-25 re-pricing based on industry rules of thumb; **sensitivity testing showed the recommendation is robust** — even at 40% retention, the projected loss ratio outcome shifts by only 0.1 points. The more material sensitivity is regulatory: if non-renewal is limited to the top 100 policies (rather than the top 249), the projected loss ratio improvement reduces from **23.3 to 19.7 points**. The recommendations remain net positive in all tested scenarios.

NEXT STEPS

With approval, I can: (1) model rate elasticity curves at multiple price-change levels to optimize Action 1's increase percentage; (2) design specific underwriting filters for Action 2 (vehicle value caps, deductible structure, prior claim exclusions); and (3) prepare the non-renewal notification documentation required for Action 3 within regulatory timelines. A phased rollout plan can be ready within two weeks.